

[DC] Enhancing Accessibility in XR Games for Users with Autism Spectrum Disorder

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ABSTRACT

Extended Reality (XR) devices, encompassing Virtual and Augmented Reality, have rapidly grown in popularity across industries such as entertainment, education, and healthcare. However, accessibility barriers remain prevalent, especially for individuals with Autism Spectrum Disorder (ASD), due to a lack of inclusive design considerations. This position paper advocates for targeted research on the accessibility of XR applications for ASD users, highlighting the importance of developing evidence-based guidelines to enhance usability and inclusion. By addressing this underexplored area, the paper aims to contribute to the creation of practical solutions that empower individuals with ASD to fully benefit from XR technologies.

Index Terms: Extended Reality, Accessibility, Autism Spectrum Disorder.

1 INTRODUCTION

With technological advances, devices related to Extended Reality, like the Meta Quest and Apple Vision Pro, have become widely available and seen a huge rise in popularity. This has led to new possibilities across various fields, like education and medicine.

Parallel to the increase of XR's availability, and in a similar timeframe, Autism Spectrum Disorder's prevalence has increased all over the world, reaching a ratio of approximately 1 in 36 children globally[3]. Autism Spectrum Disorder often results in an impairment in many aspects of social interaction, such as empathy, communication and social perception[6].

Accessibility, which is the practice of making information, activities, and/or environments sensible, meaningful, and usable for as many people as possible, is not only something which all software developers and designers should strive to achieve, but also a legally mandated practice. In the European Union, for instance, a European Accessibility Act[4] was passed, with the aim of improving the functioning of the internal market for accessible products and services, by removing barriers created by divergent rules in Member States.

Despite legal mandates like the European Accessibility Act and the demonstrated potential of XR technology to support users with autism, research on accessibility in this field remains limited. As of November 2024, no evidence-based guidelines exist to help developers create XR applications that are more accessible to autistic users.

This research aims to develop a set of evidence-based guidelines for creating accessible XR applications specifically designed for users with autism. Given the wide spectrum of ASD, the focus will be on autistic users with minimal intellectual impairment, narrowing the scope to address specific traits that manifest within this subgroup. A variation of the Design-Science Research approach[7] will be used, where instead of developing an artifact, the goal will

be to establish what the optimal sensorial and systemic aspects for players with ASD in XR are. A Human-Centered collaborative design approach[5] will be followed.

2 SIGNIFICANCE

This research contributes to the field by addressing a significant gap in our understanding of how XR applications can be tailored to meet the needs of ASD users. Studies involving ASD users and XR technology have shown overwhelmingly positive results. It's a technology that provides a versatility and a sense of presence and immersion[8] that can help users feel more involved in the tasks they perform in the application, fostering a greater sense of engagement.

Furthermore, and as already mentioned, there are no universal guidelines for the development of XR applications for ASD users. The evidence-based creation of these and the verification of their effectiveness will lead to a better understanding for developers who want to make their XR applications accessible for ASD users, expanding knowledge on XR accessibility as a whole. This means not only physical accessibility, but also cognitive accessibility and with the specific focus of ASD users.

As an added bonus, this research will provide a framework for future studies and serve as a case study for other researches interested in accessibility, XR or game design for neurodiverse populations. It could also lead to new research questions, such as how different subgroups within the autism spectrum interact with XR technologies or how similar approaches could be adopted for research of other neurodiverse conditions.

3 GOALS AND PROGRESS

The specific goals of the research are to:

- Finding and assessing the specific characteristics of the sensorial and systemic aspects [1] in an XR game that pose a difficulty to users within our profile.
- Finding existing solutions related to these aspects.
- Proposing new solutions for the most severe unsolved aspects, and migrating existing solutions to our user profile, thus filling the identified gaps.
- Validating the proposed solutions through quantitative and qualitative evaluation methods.

This will be achieved by following an adapted version of the Design Science Research (DSR) paradigm, integrating a Human-Centered participatory design approach. The paradigm consists of six main phases, called activities[7]:

- 1. Identify Problem & Motivate:** A state of the art will be performed through an exploratory literature review on the topic of XR games and autism and through a contextual observation of users to specify exactly what are the aspects that pose a difficulty to users within the defined profile.
- 2. Define Objectives of a Solution:** The solution developed will not be a piece of technology, but rather a set of instructions to be used for people who develop technology in XR for people with ASD. In this activity the priorities, limits and scope of the solution will be defined.

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3. **Design & Development:** The set of instructions will be developed following best practices and state of the art methodologies.
4. **Demonstration:** In the virtual reality environment different combinations of the several aspects mentioned in the developed instructions will be built. The focus will be set on the aspects defined in Activity 3. The virtual environment will be created using the Unity software as a graphic motor, due to its capabilities and high productivity. The different combinations created in the virtual environment will be evaluated by a user test.
5. **Evaluation:** For this activity, an appropriate set of metrics or theoretical framework will be researched to find out how correct are the proposed solutions.
6. **Communication (publications):** 2 systematic literature reviews will be published on the subject, as part of the state of the art from Activity 1. Afterwards, an article related to VR and personalization is to be published, as well as the papers regarding the preliminary results of the experiment and the final suggested solutions once it has been thoroughly tested.

Currently, this project has already published a systematic literature review as part of the state of the art outlined in Activity 1, focusing specifically on the engineering aspects of the issue. It contains a set of 12 recommendations extracted from the analysed articles, such as reducing the sensory load and using auditive and visual prompts to complement textual communication [2]. A second literature review, which takes a medical perspective, has been completed but remains unpublished for now.

Additionally, a basic VR environment has been created using the game engine Unity 3D. This environment will be used to test some of the recommendations extracted in Activity 1, and defined and curated in Activities 2 and 3.

4 DISCUSSION

While the purpose of this research is clear, several counterarguments can be made that challenge its validity and scope:

- Autism is, by definition, a spectrum, with individuals exhibiting a wide range of needs, preferences and symptoms. This diversity can affect the research in two ways. Firstly, attempting to establish a set of evidence-based guidelines that encompass the needs of all ASD users is overly ambitious and risks the guidelines becoming too broad to be useful to anyone. Secondly, the limited amount of users we will likely have as a sample can lead to a skewed research where the solutions found are too specific for the general ASD population.
- Conducting usability testing in users with ASD can be challenging due to communication barriers and extreme sensory sensitivities, specially when using a VR headset. Not only because of physical discomfort, but also because of cognitive barriers. These issues also raise ethical concerns, such as obtaining informed consent from users, properly managing participant expectations and ensuring the study causes no stress or harm to the participants of the experiment.
- As of November 2024, there is a lack of Universal Standards in accessibility design for ASD, complicating the design of features that can be broadly effective across the spectrum.

However, these counterarguments can be addressed and rebutted in the following ways:

- While autism is indeed a broad spectrum, we explicitly acknowledge this challenge and propose narrowing our focus to a specific subgroup. This targeted approach ensures that the solutions we develop are both practical and meaningful, even if they may not be universally applicable. This also helps mitigate the risks that arise from having a small sample size.
- The absence of universal standards highlights the need for research in this area. By collaborating closely with ASD users and relevant associations, we aim to provide empirical evidence and actionable insights that can inform the development of these standards.
- While usability testing with ASD users presents challenges, there is substantial evidence showing that these obstacles can be overcome through careful planning — such as familiarizing users with the environment — and adaptive methods, including shorter sessions, reduced sensory input, and close collaboration with caretakers and families[2].

5 FINAL REMARKS

In conclusion, the increasing prevalence of ASD and the widespread adoption of XR technologies highlight the urgent need for research into making XR applications accessible to users with ASD. While challenges such as the broad nature of the autism spectrum and the lack of universal accessibility standards remain, this position paper has outlined how targeted, evidence-based guidelines can contribute to more inclusive design practices. By focusing on a limited group and collaborating with the community, we aim to offer practical, actionable solutions for developers. This research seeks not only to address current gaps in the literature but also provide a foundation for future studies, helping to shape a more inclusive and accessible XR landscape for neurodiverse populations. Through ongoing work, the development of these guidelines and their validation will improve our understanding of how XR technologies can be tailored to support ASD users.

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